Issue
011

Reports on ICON

CAMS 3D climatology and forecasted aerosols as an option in ecRAD radiation in ICON

Harel Muskatel, Daniel Rieger, Pavel Khain, and Martin Köhler

August 2024

DOI: 10.5676/DWD_pub/nwv/icon_011
ISSN: 2628-4898

Deutscher Wetterdienst
Wetter und Klima aus einer Hand

Max-Planck-Institut
für Meteorologie

DOI: 10.5676/DWD_pub/nwv/icon_011
ISSN: 2628-4898

The CC license “BY-NC-ND” allows others only to download the publication and share it with others as long as they credit the publication, but they can’t change it in any way or use it commercially.

Publisher

Deutscher Wetterdienst
Business Area “Research and Development”
Frankfurter Straße 135
63067 Offenbach
www.dwd.de

Editors

Sebastian Rast, MPI-M
sebastian.rast@mpimet.mpg.de
René Redler, MPI-M
rene.redler@mpimet.mpg.de
Daniel Reinert, DWD,
daniel.reinert@dwd.de
Daniel Rieger, DWD,
daniel.rieger@dwd.de
Florian Prill, DWD,
florian.prill@dwd.de

CAMS 3D climatology and forecasted aerosols as new options in ecRAD radiation

Harel Muskatel (1), Daniel Rieger (2), Pavel Khain (1), Martin Köhler (2)

(1) Israel Meteorological Service,
(2) Deutscher Wetterdienst

This document refers to release icon-2024-07.

Abstract

The Copernicus Atmosphere Monitoring Service (CAMS) 3D climatology and forecasted aerosols are now implemented in ICON model as an input to the ecRAD radiation scheme. For more than a decade CAMS model was used as a reliable source of air quality forecasting system around the globe. These kinds of models require huge computational resources. Using their products as an "offline" model is a reasonable intermediate solution between the costly fully-coupled aerosol model and the cheaper aerosols climatology. These forecasted aerosols are interpolated in space and time and treated as a background to the radiation scheme (no internal advection inside ICON). The second option which is now implemented in ICON, is the CAMS 3D climatology. It has several advantages over the Tegen 2D climatology as explained in the introduction. This report contains the description of how to run ICON using each of the new developments: the 3D CAMS climatology (irad_aero = 7) and the CAMS forecasted aerosols (irad_aero = 8).

Keywords: Aerosols, Radiation

Correspondence to: Harel Muskatel (muskatelh@ims.gov.il)

1. Introduction

Aerosols in the atmosphere have a significant impact on weather and climate. They directly interact with both incoming solar radiation and outgoing thermal radiation through absorption and scattering. Furthermore, they are necessary for cloud formation which effects not only the radiative fluxes but also precipitation and other atmospheric processes. The direct effect on radiation can be easily seen in dust outbreaks and volcanic ash eruptions, but even on regular conditions aerosols concentration can have large day-to-day variability imposing significant errors when climatological averages are used. For many years, many NWP models including ICON (Zängl et al., 2015), used the Tegen 2D monthly climatology (Tegen et al., 1997). In this dataset, the total column AOD of five generalized species of aerosols is available for one wavelength (550 nm). This single number is used to evaluate the optical properties at each vertical level assuming some exponential vertical density profile. Correction factors are then used to calculate these properties to other model's spectral intervals. These assumptions (i.e. constant size distr. and

chemical composition, no hygroscopical growth) are not needed when using the CAMS 3D aerosols. The Copernicus Atmosphere Monitoring Service (CAMS) is an ECMWF project, which provides global air quality analysis and forecast including aerosols and other tracers (Morcrette et al., 2009). The model is based on IFS and has a coarser spatial resolution but with the same vertical resolution (137 levels). The output forecast lead time is 5 days with a 3-hours temporal resolution. 3D mixing ratios of 11 aerosols tracers are calculated: 3 size bins of mineral dust, 3 size bins of sea salt, sulphate, hydrophobic and hydrophilic tracers of both black carbon and organic matter. The optical properties were pre-calculated by ECMWF for all ecRAD's thirty spectral intervals. Another advantage is that the optical properties for the hydrophilic aerosols are relative-humidity dependent. CAMS climatology (A. Bozzo et al., 2017) is based on the reanalysis (2003-2014) of CAMS model for the mentioned aerosol species in a 60-levels vertical resolution. In this case, the vertical profiles are a result of actual dynamics resulting a monthly climatology of aerosols profiles (A. Bozzo et al., 2017). The dataset contains monthly averages of layer integrated mass from which the 3D mixing ratios are calculated and used as an input to ecRAD scheme.
We performed several tests using both schemes in regional and global scales and the direct effect on radiation was evaluated with positive results. In the Israel Meteorological Service (IMS), the operational set up since August 2023 uses CAMS climatology as default aerosols, and the near future plan is to switch to CAMS forecasted aerosols when dust outbreaks are foreseen. On the regional scale, no significant increase in runtime was detected compared to the previous Tegen climatology input. In terms of memory, two time-levels of 11 additional 3D variables are needed. The CAMS 3D aerosol data can be further used for cloud ice and cloud droplets nucleation parametrizations. These parametrizations are still under development.

2. Pre-processing

This section briefly outlines how CAMS climatological and forecasted data can be prepared for the usage in ICON. For running ICON with CAMS climatology (`irad_aero = 7`):

- The original global dataset from https://confluence.ecmwf.int/display/ECRAD should be downloaded. You can find the '3D monthly aerosol climatology derived from CAMS reanalysis system' in the 'Other resources' section.
- The file should be renamed to `aerosol_cams_3d_climatology_2003-2013_orig.nc`.
- The script `scripts/preprocessing/make_camsclim_onICONgrid.sh` should be used in the same directory as the grid and data files.
- Running this script will create the CAMS climatology for your ICON grid and domain using CDO (needs CDO version 2.1.0 and above).
- The script also contains the description on how to run it.
- The only editing needed in the script is the `GridName`. For example: if the grid file is: `R02B06_DOM01.nc` then `GridName=R02B06_DOM01` should be set.
- Executing the script will do the rest.
- The comments at the end of script provide some hints in case of errors during the execution of the script.
- In this example, the resulting CAMS climatology file will be: `icon_cams_clim_R02B06_DOM01.nc`
- In the ICON namelist file, the path to this climatology file should be: `cams_aero_filename='icon_cams_clim_R<nroot0>B<jlev>_DOM<idom>.nc'`

For running ICON with forecasted CAMS aerosols (`irad_aero = 8`):

- The user needs to download CAMS data for the 11 aerosol species `aermr01, ..., aermr11` from the MARS database (or from ECMWF FTP server). The data should be downloaded to one NetCDF file which contains all aerosols and timesteps together.
- This file should be named `$CAMSdir/CAMS_aero_${dathh}.nc` where `$CAMSdir` is the location of the CAMS files and `$dathh` is the date and hour of the initial forecast time in the format `yyyymmddhh`.
- The 3D pressure data of the CAMS forecast is also needed. Unfortunately, this data is not explicitly available in the model output and needs to be calculated. Therefore, the 2D data of the natural logarithm of surface pressure `lnsp` needs to be downloaded as well. The `lnsp` dataset should be downloaded/provided in `grib` format and in separate files for each timestep. Naming should be in the format:
  `$CAMSdir/CAMS_lnsp_${dathh}_${n}.grb`
  where "n" is the forecast hour.
- The 3D pressure is calculated by the python script
  `scripts/preprocessing/compute_pressure_on_ml.py`
  which is based on:
  `compute_geopotential_on_ml.py`[1]
  `CDO`, `python3` and `ecmwf-toolbox` is needed for this script to work.
- The script for this purpose is located in
  `scripts/preprocessing/make_camsforc_onICONgrid.sh`
  The naming convention and the name-list treatment should be the same as described above for the CAMS climatology.

3. Vertical interpolation

The values of aerosols mixing ratios are needed for the direct aerosols-radiation interactions calculations in ecRAD. However, the CAMS 3D climatology dataset contains layer-integrated mass ($\frac{\mathrm{kg}}{\mathrm{m}^2}$) for each aerosol type and also the half-level pressure for each layer (60 levels). The CAMS climatology has a different number of pressure levels from ICON and also different lowest/highest pressure values. In order to conserve the total column aerosol mass of the original CAMS data and to reasonably distribute the mass within ICON levels, sigma coordinate in both models is used. The code loops on each ICON sigma level and calculates the amount of layer-integrated mass that is confined within it. For each ICON sigma layer, the amount of mass confined inside it is calculated using pressure based linear interpolation. Three cases are possible: ICON layer is completely within one CAMS layer, ICON layer covers two CAMS layers and ICON layer covers a few CAMS layers (see Figure 1). The mixing ratios ($\frac{\mathrm{kg}}{\mathrm{kg}}$) for each ICON level are then calculated by multiplying layer-integrated mass with the gravitational constant divided by pressure thickness.
The CAMS forecasted data, on the other hand, is given as mixing ratios ($\frac{\mathrm{kg}}{\mathrm{kg}}$) for 137 vertical levels. For the same reasons mentioned above: different pressure boundaries, different number of layers and mass conservation, the data is first translated to layer-integrated mass. Then, the same procedure as described above for the CAMS climatology is executed.

[1] https://confluence.ecmwf.int/display/CKB/ERA5%3A+compute+pressure+and+geopotential+on+model+levels%2C+geopotential+height+and+geometric+height

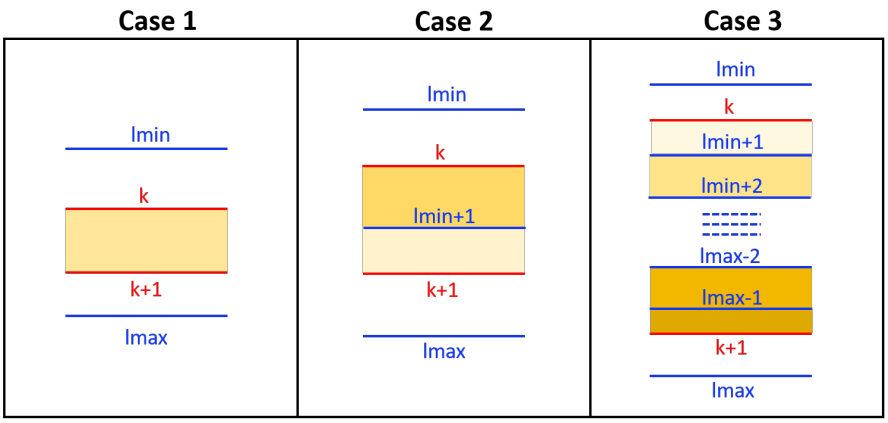

Figure 1: Vertical interpolation of layer integrated mass ($\frac{\text{kg}}{\text{m}^2}$) from CAMS (blue) to ICON (red) levels. The algorithm loops over all `k` ICON levels and searches for the CAMS levels lmin and lmax which are below and above `k` and `k+1` levels respectively. The mass confined inside the layer between `k` and `k+1` levels is then calculated. Three cases are possible:
Case 1: ICON layer between `k` and `k+1` is completely within one CAMS layer. In this case, only the relative proportion of the mass in the layer between `k` and `k+1` is taken into account (yellow layer). The mass is calculated by multiplying CAMS layer integrated mass between `(lmin, lmax)` by $\frac{P_{k+1}-P_k}{P_{lmax}-P_{lmin}}$ where P is the pressure in sigma coordinates.
Case 2: ICON layer covers two CAMS layers. Here we include the mass confined between `(k, lmin+1)` proportionally taken from the CAMS layer `(lmin, lmin+1)` and the mass confined between `(lmin+1,k+1)` proportionally taken from the CAMS layer `(lmin+1, lmax)`.
Case 3: ICON layer covers a few CAMS layers. As in case 2 but now adding all masses confined in the CAMS layers which are confined between `(k, k+1)`.

4. Results

Limited area model runs on the IMS domain ($25-39°\text{E}, 26-36°\text{N}$) showed positive results with both CAMS climatology and CAMS forecasted aerosols. ICON with forecasted CAMS input was tested for the entire 2020 year in the IMS domain and reduced both bias and RMSE of global radiation compared to Tegen climatology input. This model especially outperforms other schemes in polluted cases. As an example, we compared the results of ICON using the Tegen climatology with the results using CAMS forecasted/climatology aerosols in April 2022. In Figure 2, the daily average AOD values in Tel-Aviv are presented for this polluted period. One can see how CAMS predictions successfully capture the day-to-day variability as was measured in AERONET station in Tel-Aviv University. The improvement in ICON radiation forecast when using CAMS forecasted aerosols is shown in the upper and lower panel of Figure 3.

ICON with CAMS climatology was also tested over 6-months period in 2022 for several locations in Israel. The model reduced surface radiation biases to less than 1 $\frac{W}{m^2}$ while other versions of ICON resulted positive biases larger than 25 $\frac{W}{m^2}$. These results led to the operationalization of the new schemes in the IMS. In this documentation we also show the results of global runs for two months period: January and July 2022. In the figures 4, 5, 6 and 7 the differences between the

CAMS climatology (left) and CAMS forecasted aerosols (right) to the default (Tegen climatology) ICON runs are shown. Top row panels show the averaged values for January 2022 and the bottom for July 2022. Figure 4 shows the top of atmosphere (TOA) net shortwave radiation (SW) fluxes differences. From this figure we can see the same patterns for both CAMS climatology and CAMS forecasted aerosols but with different magnitudes. There is less TOA SW radiation in the Saharan desert and the Arabian Peninsula due to higher dust AOD compared to the Tegen climatology (see e.g. Figure 4 in A. Bozzo et al. (2017)). On the other hand, more radiation related to lower AOD values can be observed in South Africa, South and North America and in Europe. The later can be explained by the anthropogenic AOD reduction that occurred between the 1980s and the 2010s decades. Figure 5 is the same but for the surface fluxes. Figure 6 also shows surfaces fluxes differences but for the longwave radiation (LW) and in Figure 7 the surface temperature differences are displayed. In winter the most significant sensitivities can be seen in the Saharan desert, Australia and North America with higher magnitudes for the CAMS forecast compared to CAMS climatology. The differences between the models are more profound in the summer especially in the Sahara and in Asia.

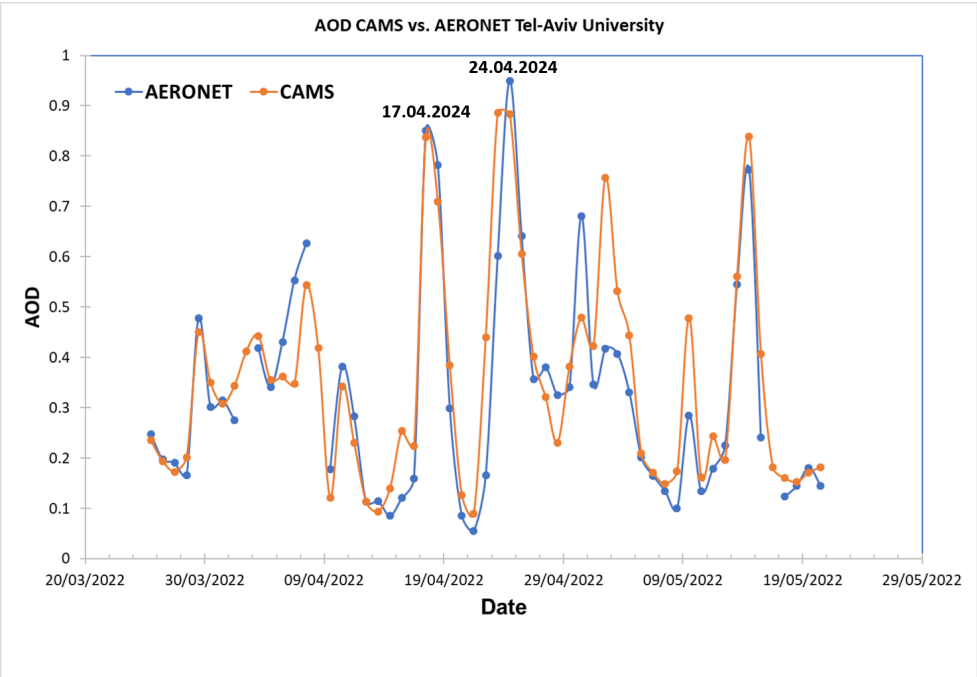

Figure 2: Measured AERONET vs. CAMS model AOD for April-May 2022 in Tel-Aviv University in Israel (data taken from `https://global-eqc-server.atmosphere.copernicus.eu/`). Notice the AOD peaks on the 17th and 24th of April 2022 due to dust events.

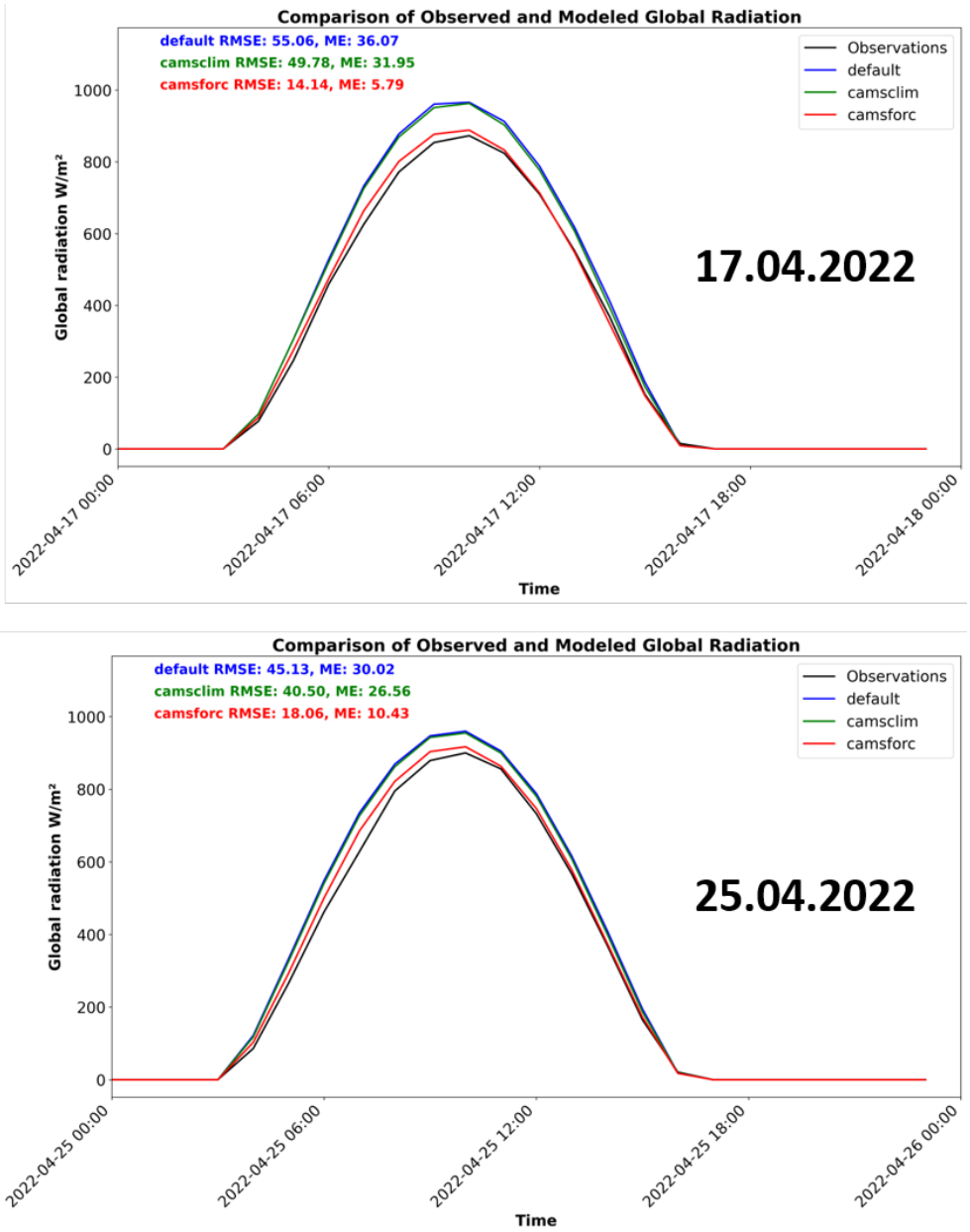

Figure 3: Global radiation verifications for the 17th and 25th of April 2022. These days were chosen due to cloudy skies conditions in other dates at this period. Cloudy skies conditions are less suitable for our purpose of evaluating aerosols effects on radiation. Black lines represents the observations in Bet-Dagan IMS radiation station near Tel-Aviv. Colored lines represents the model surface radiations results for Bet-Dagan's nearest grid point when running ICON using 3 aerosols inputs: default (Tegen) in blue, CAMS climatology (green) and CAMS forecasted aerosols (red). The radiation reduction caused by the dust is successfully forecasted by the ICON-forecasted-CAMS version (red). The reduction of mean error (ME) and RMSE can also be read in the upper-left corner of the plots. The model runs were initiated every day at 00:00UTC but only the 24-48 hours lead-time were included in the verifications. We also evaluated the results for the whole period of 16-27 of April 2022: the RMSE values were 75, 70 and 64 $\frac{W}{m^2}$ and the ME values were 15, 12 and 6 $\frac{W}{m^2}$ for these three setups respectively (default, cams-clim, cams-forc).

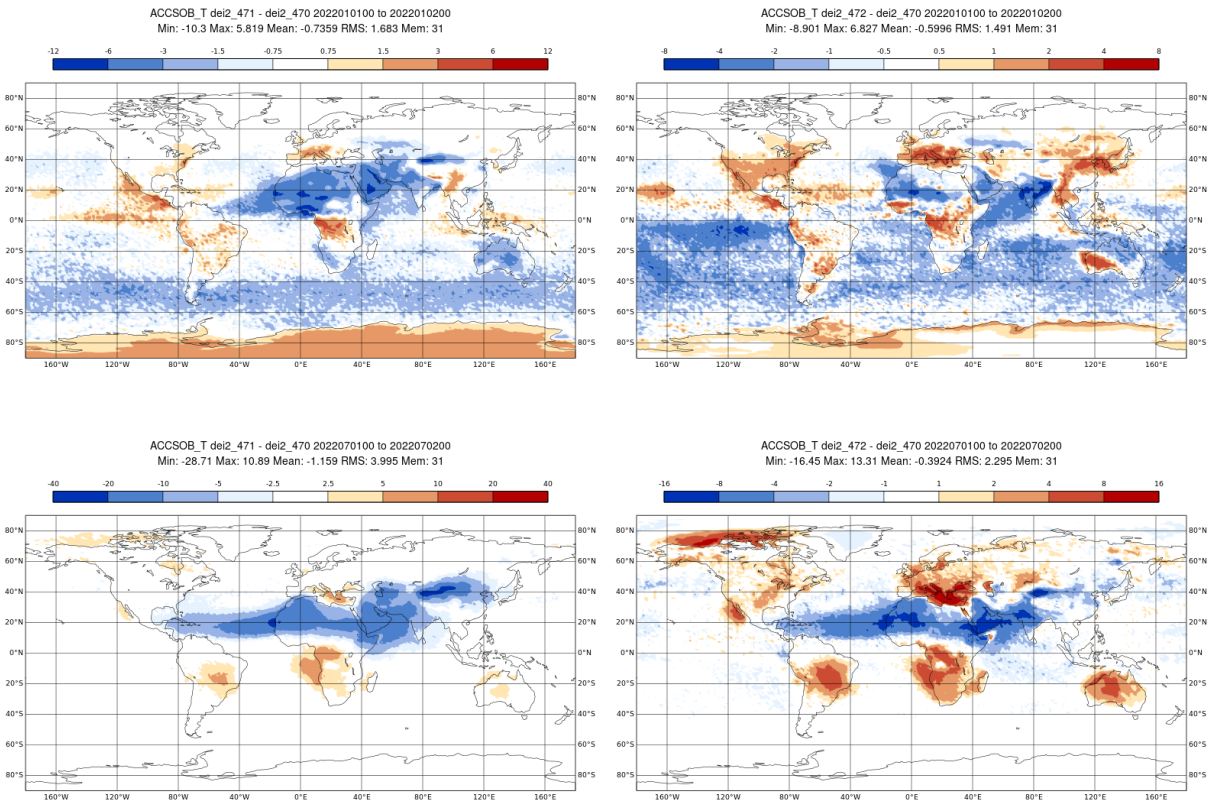

Figure 4: TOA net SW radiation fluxes differences. Top: January 2022. Bottom: July 2022. Left: CAMS climatology vs. Tegen climatology. Right: CAMS forecasted aerosols vs. Tegen climatology

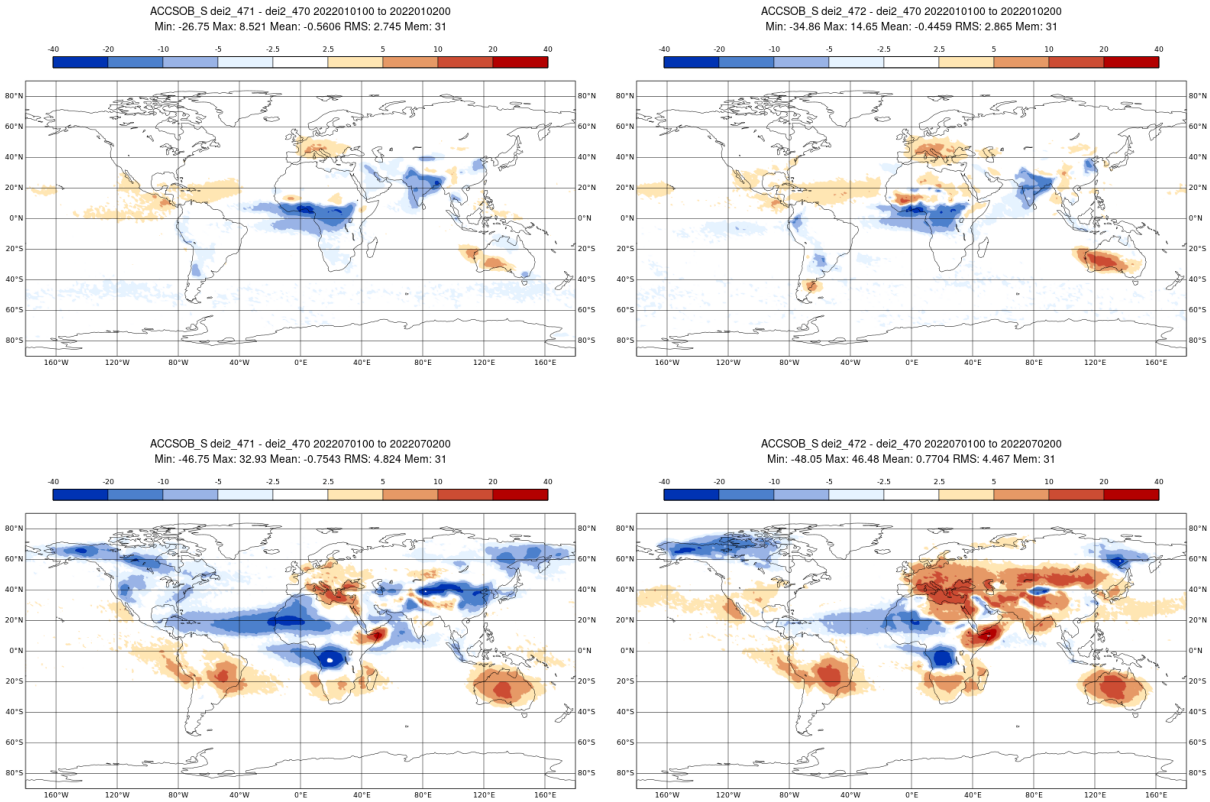

Figure 5: Same as Figure 4 but for Surface SW radiation

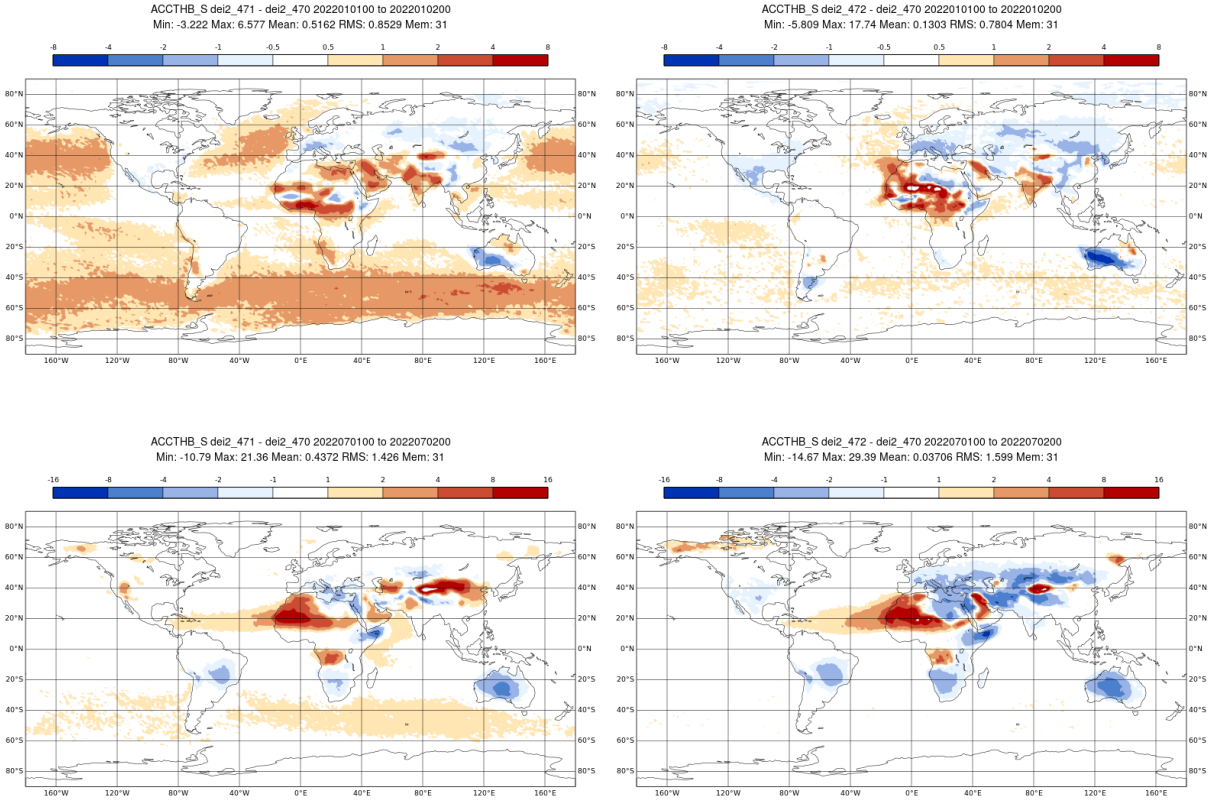

Figure 6: Same as Figure 4 but for Surface LW radiation

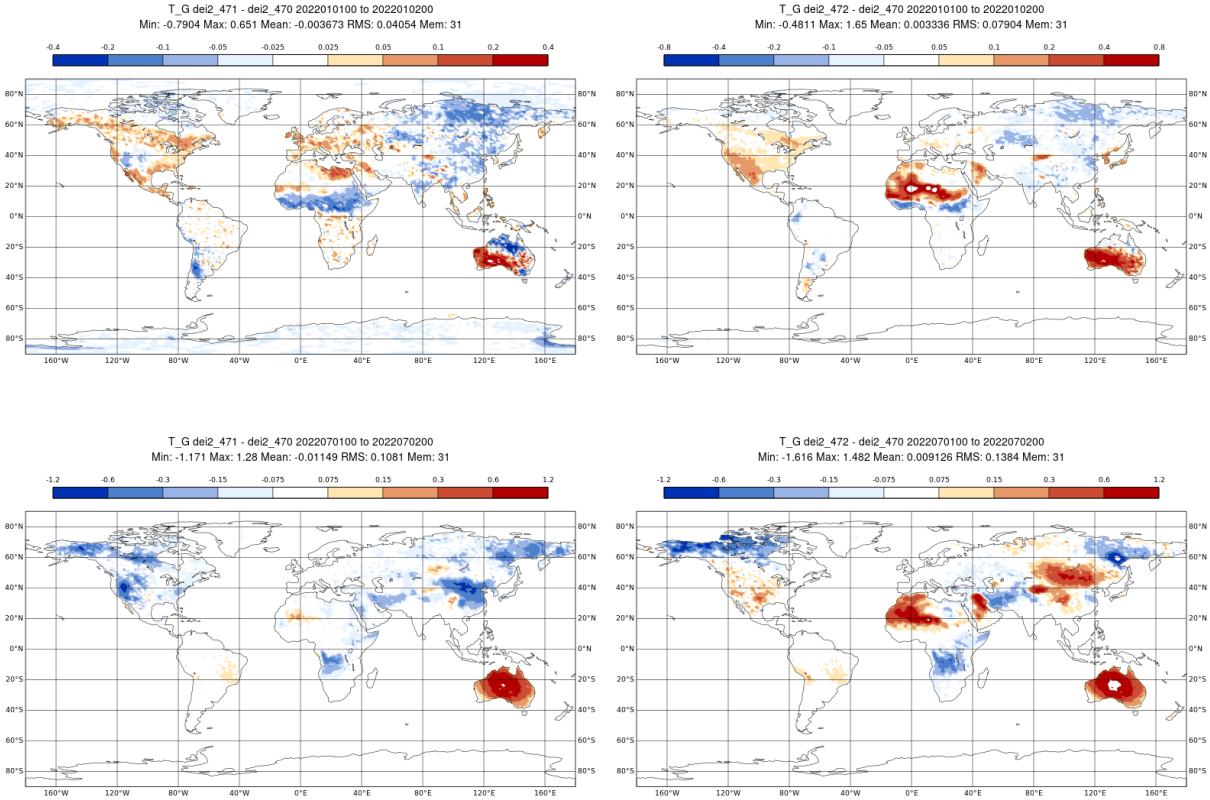

Figure 7: Same as Figure 4 but for Surface temperature

5. Conclusions & Outlook

In this work we introduced the up-to-date CAMS 3D aerosols climatology and the CAMS forecasted aerosols as an input to ICON radiation scheme. These are more up-to-date alternatives to the 2D Tegen climatology used in ICON for many years. The advantages were discussed and some preliminary results were presented. The two new options can be used on global or regional scales with the usual model tuning needed upon setup changes. Further research can be done on the usefulness of CAMS forecasted aerosols in PV power output forecasts, especially during dust outbreaks. The ICON-CLM community can examine the CAMS 3D climatology impact on climate model runs. For the first time, the vertical profiles of aerosols with their significant seasonal dependency are modeled. Aerosols-cloud interactions are also important aspect which need further investigation. Therefore, our next step will be implementing these 3D CAMS aerosols as an input to ICON cloud droplets activation scheme.

References

A. Bozzo, S. R., A. Benedetti, J. Flemming, P. Bechtold, M. Rodwell, and J.-J. Morcrette, 2017: Implementation of a CAMS-based aerosol climatology in the IFS. *ECMWF Technical Report*, **801**.

Morcrette, J.-J., and Coauthors, 2009: Aerosol analysis and forecast in the European Centre for Medium-Range Weather Forecasts Integrated Forecast System: Forward modeling. *J. Geophys. Res.*, **114**.

Tegen, I., P. Hoorig, M. Chin, I. Fung, D. Jacob, and J. Penner, 1997: Contribution of different aerosol species to the global aerosol extinction optical thickness: Estimates from model results. *J. Geophys. Res.*, **102**, 23,895—-23,915.

Zängl, G., D. Reinert, P. Ripodas, and M. Baldauf, 2015: The ICON (ICOsahedral Non-hydrostatic) modelling framework of DWD and MPI-M: Description of the non-hydrostatic dynamical core. *Q. J. R. Meteorol. Soc.*, **141**, 563–579.